PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: Balancia

Version:

1. Project Overview

- **Problem Statement:** Managing employees and tasks in small to medium businesses is time-consuming and inefficient. Managers and business owners not only have to micro-manage employees, but also assign tasks manually. Consequently, this leads to inefficient task allocation and reduced productivity.
- **Target Domain:** AI-powered workforce management system for small to medium-sized businesses by focusing on analyzing and automating tasks, tracking productivity, and making the most cost-efficient decisions.
- **Proposed Solution Summary:** Balancia uses z.AI to help managers and business owners to manage their teams more efficiently. They can input vague goals, and the AI will analyze and automatically break them into tasks to be assigned to suitable employees. It also tracks the employee's progress and productivity and prevents overworking, This allows managers and business owners to shift from micro-managing to making strategic decisions.

2. Background & Business Objective

- **Background of the Problem:** In many organizations, managers and business owners spend a large amount of time on micromanagement, which reduces overall productivity. Research shows that 79% of managers admit that they need to micromanage their teams regularly. This consumes time away from more important strategic decisions and slows down the overall workflow. In fact, micromanagement is estimated to cost companies up to $398 billion annually in lost productivity and employee turnover (Yaware, 2025). This shows that micromanagement is inefficient and can create serious financial and operational impacts for organizations.

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- **Importance of Solving This Issue:** AI adoption can improve productivity. Studies indicate that AI tools can increase productivity by around 20% to 40% in tasks such as coding and knowledge work (SoftwareSeni, 2025). In addition, surveys show that workers can save 40–60 minutes per day using AI tools (Tom's Hardware, 2025). However, without proper systems, these benefits are not fully realized. Around 40% of time saved by AI is lost due to rework and corrections (ERP Today, 2026). Hence, there is still a need for a smarter system that not only uses AI but also analyzes workflows and automates task allocation.
- **Strategic Fit / Impact:** This system aligns with the global shift towards AI-driven work environments. Research shows that AI could generate up to $2.9 trillion in economic value annually by transforming how people and intelligent systems work together (McKinsey Global Institute, 2025). Besides, 20%–40% of workers are already using AI tools in their jobs (Federal Reserve, 2025). This demonstrates a strong trend towards integrating AI into daily operations. Hence, this system supports business owners and managers to become more efficient and competitive.

3. Product Purpose

3.1. Main Goal of the System: To provide an AI-powered management assistant that improves productivity and reduces micro-management.

3.2. Intended Users (Target Audience):

- Founders of small to medium businesses
- Project Managers
- Entrepreneurs
- Startup teams
- Team leaders managing employees

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4. System Functionalities

4.1. Description:

The system operates as an AI-powered management engine that transforms vague business goals into structured workflows. As Balancia uses natural language input, the AI analyzes goals, identifies required tasks, then matches them with suitable employees, and continuously tracks progress and productivity. The system integrates data analysis and task management and AI decision-making into a single platform to improve efficiency and reduce manual coordination.

4.2. Key Functionalities:

- **Goal Decomposition:** Transforms vague user inputs (e.g. "launch a website") into structured, step-by-step tasks using the GLM model.
- **Smart Task Allocation:** Automatically assigns tasks to employees based on their profile, including skills, availability, performance data, and personal attributes (e.g. MBTI, career aspirations).
- **Overwork Detection:** Detects when employees are overloaded based on working hours and task load, and alerts managers to redistribute tasks.
- **ROI Analytics:** Provides analytics on productivity trends, team capacity, and performance to help managers evaluate the impact of AI-driven decisions.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: Balancia uses the Z.ai GLM as the core AI model. This model is chosen because it has strong natural language understanding and reasoning capabilities which helps in transforming vague goals into structured tasks. It is also suitable for multi-step reasoning as it can let the system analyz user input, then identify required skills, and generate task allocations efficiently. The model also supports API integration so that it is practical for real-time system deployment.

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4.3.2. Prompting Strategy: Balancia uses a multi-step prompting strategy. First, the model interprets the user's vague goal and extracts the key objectives. Then, it breaks the goal into structured tasks. Finally, it matches tasks with employees based on skills and availability.

This approach is chosen because a single-step prompt may produce incomplete or inaccurate results. Multi-step prompting improves accuracy, consistency, and reasoning quality, which is important for task planning and allocation.

4.3.3. Context & Input Handling: The system handles unstructured input by processing natural language goals and converting them into structured data. To manage large inputs and reduce cost, prompt compression and filtering are applied. Instead of sending full employee data, only relevant skill-related information is included.

The system also applies limits on input size. If the input exceeds the maximum allowed length, it will be split into smaller chunks before processing. This ensures stable performance and prevents excessive token usage.

4.3.4. Fallback & Failure Behavior: When the AI generates off-topic or even low-quality outputs, the system applies several safeguards. First, responses are validated before use. If the output is not usable, the system can retry with adjusted prompts.

Besides, all AI-generated task plans must be reviewed and approved by the admin before execution. This human-in-the-loop approach ensures accuracy and reduces risk. If repeated failures occur, the system will display a fallback message and allow manual task creation.

5. User Stories & Use Cases

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- **User Stories:**

- **As a project manager,** I want to input vague goals in natural language so that AI distributes tasks to the right employees based on their skills and availability as to why the employees are suitable for these tasks. I also want to track the progress of the project and employee workloads in real-time so that I don't accidentally overwork anyone. I want AI to detect when the employees are stuck and suggest help so that the project does not become delayed and the employees do not become overstressed. Finally, I want to see monthly/yearly ROI analysis to see the impact of the AI decisions on our company.
- **As an employee**, I want to receive the tasks that align with my skills, preferences and energy peaks so that I can work more efficiently. I want AI to notice the time I am struggling so that I can get support without asking for help in person. I want an unbiased record on my efficiency and productivity so that I get more opportunities in my career based on my actual performance data.

- **Use Cases (Main Interactions):**

- **Vague Goal to Task Allocation Flow**: The manager enters the goals dashboard, clicks the "Create Goal & Generate Tasks" button, and then inputs vague goals in natural language. Balancia uses GLM to parse the intent and transform it into structured information, identify the required skills and components, and scan the employee table for checking detailed skills and calibration (MBTI/Aspirations) to display assigned tasks to the employees with deadlines.
- **Tracking real-time progress:** Balancia tracks real-time progress of the projects, all team tasks, and employee performance across the different related dashboards for the manager. When the manager clicks the goals dashboard, it will display active goals, their progress, and velocity. When the manager clicks on each project, it will show the allocated tasks for its project. When visiting the "Employee" and "Tasks" dashboards, the system will show detailed related information.
- **Automated Overwork Detection**: The system triggers alerts based on real-time tracking and gives notifications to the user. When an employee exceeds the average weekly

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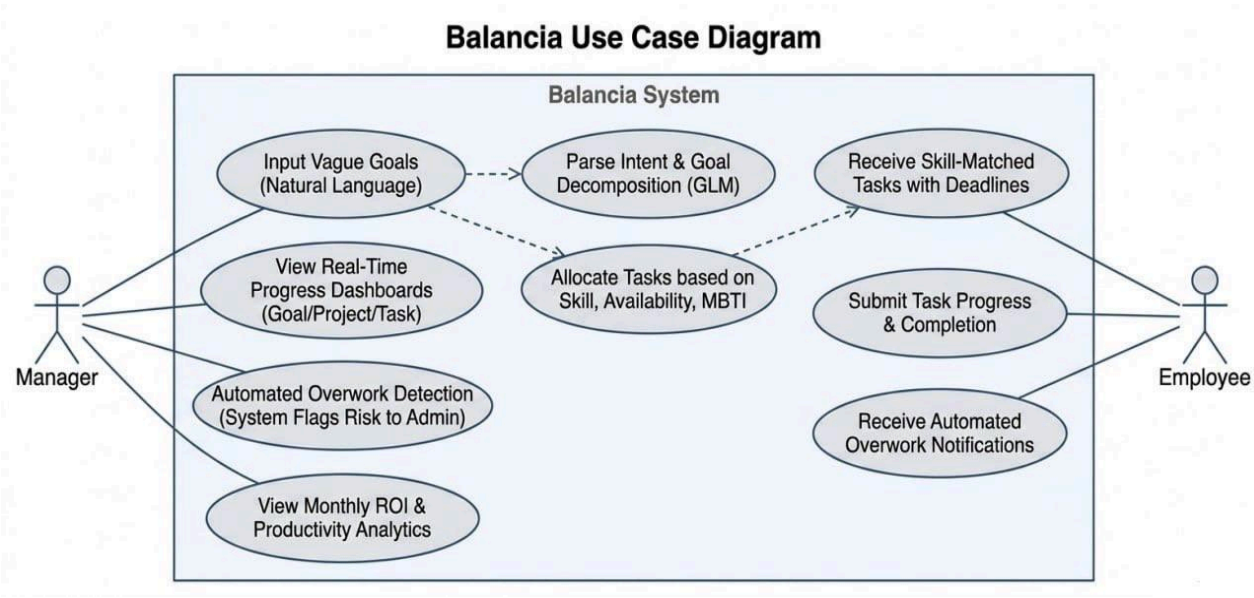
working hours, the system identifies them as an overwork risk and lets the admin know to redistribute tasks. Additionally, if a project still requires significant progress but is approaching its due date, the system generates an alert indicating that the project is at risk.

- **Monthly ROI & Productivity Analytics:** The manager can review the "Analytics & Insights" dashboard to consider the impact of the AI-driven decision. Balancia visualizes monthly productivity & hours trends, individual performance breakdowns, and task status distribution. It compares average productivity against the previous month. It displays the percentage of the team capacity(e.g, 92%), highlights the overwork risk, and prompts the manager to "Redistribute Tasks" to maintain employees' work-life balance.

- **Use Case Diagram:**



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6. Features Included (Scope Definition)

- **Goal Decomposition (Natural Language Processing):** Allows admins to input vague business goals, which are automatically transformed into structured, step-by-step tasks using the GLM model.
- **Smart Task Allocation:** Assigns tasks to the most suitable employees based on their skills, availability, and personal attributes (e.g., MBTI, career aspirations).
- **Productivity Dashboard:** Provides a centralized view for admins to monitor employee productivity and task progress or status (e.g., completed/pending).
- **Overwork Detection:** Detects when employees exceed a healthy workload and notifies admins to redistribute tasks.
- **ROI Insights:** Generates insights like productivity trends, team capacity, task distribution, and performance analysis so the manager can focus on decision-making.

7. Features Not Included (Scope Control)

- **Employee Messaging System:** Real-time chat between employees is excluded from the MVP to keep the focus on core AI task orchestration.
- **Mobile Application:** The system is currently web-based only. Mobile app development is planned for future versions.
- **Dark Mode:** Only a light theme is supported in the MVP to maintain simplicity and consistency. Additional UI features will be added later.
- **Personalized AI Coaching:** Advanced features such as individual skill development, coaching, or career-path recommendations are not included in this version.
- **Multi-language Support:** The system currently supports English only. Support for additional languages will be added in future updates.
- **Third-party Integration**
  Direct integration with tools such as Microsoft Teams, or Trello is not included in the MVP but is planned for future development.

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8. Assumptions & Constraints

- **LLM Cost Constraint:** Based on how much work it takes the AI to turn a simple sentence into a detailed task list, we estimate each session will cost about [INSERT TOKEN AMOUNT] tokens.

  To reduce our API cost, we use Prompt Compression. Instead of sending our entire "Employee Database" (which is huge and expensive) to the AI every time, we only send a "Filtered Vector." This means we only show the AI a small "cheat sheet" of the specific skills needed for that one goal. This keeps the Context Window small, making the AI much faster and cheaper to run at scale.
- **Technical Constraints:** The Balancia dashboard is a Web-only platform built with React/Next.js. This means it is optimized for desktop browsers and does not currently support Native Mobile features like phone notifications or iOS widgets. For data, you can export your analytics and tasks as CSV or PDF files. However, direct integration with other software, such as Jira, Asana, or Payroll systems, is not supported in this initial version.
- **Performance Constraints:** Creating a full project roadmap with skill-matching logic takes the AI about 5–12 seconds because it has to do "deep reasoning" to get the tasks right. To keep the system stable and prevent the servers from slowing down, we have set a "Reasoning Limit". Additionally, each manager is limited to 5 goal generations per hour to ensure a smooth experience for everyone.
- **User Input:** The AI acts only as a "suggestion engine", meaning it cannot automatically assign tasks to employees on its own. To ensure accuracy and safety, every project roadmap must be reviewed and manually approved by an admin before any tasks are officially sent out.

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9. Risks & Questions Throughout Development

- **Design Similarity: How are we ensuring that AI is not generating similar designs across users?**

Approach: The system reduces similarity by using unique contextual data such as project goals, employee skills, workload, and team composition. This ensures that each generated task plan is customized instead of repetitive.

- **Frame Stability: Does the user need to fix the layout if it is found broken after exporting?**

Approach: The system organizes tasks into clear hierarchies and uses an adaptive dashboard layout so the readability is clear. But, we will still make minor manual adjustments for highly complex projects.

- **Drawing Interpretation: What will happen if the AI is unable to contextualize the sketch that is provided?**

Approach: If the AI cannot accurately interpret the input, the system will prompt the user for clarification or regenerate the task plan. Moreover, all outputs require the admin to review before execution to ensure correctness and reduce errors.

- **Task Logic Personalisation: How do we ensure the AI doesn't give every manager the same "cookie-cutter" plan for every project?**

Approach: We ensure the AI prioritises unique data points, such as the specific individual skill levels and current workload of the actual team members, so that every roadmap is custom-built for that specific group.

- **Dashboard Stability: Will the user need to manually fix the dashboard if the AI generates too many tasks or very long descriptions that overlap?**

Approach: We are developing a "Smart Canvas" that automatically resizes cards and timelines. However, we are still testing whether a human will need to "clean up" the view if a project becomes extremely complex.

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10. Reference

ERP Today. (2026). *AI productivity gains are lost to rework*. Retrieved from https://erp.today/workday-research-finds-ai-productivity-gains-are-lost-to-rework/

Federal Reserve. (2025). *Measuring AI uptake in the workplace*. Retrieved from https://www.federalreserve.gov/econres/notes/feds-notes/measuring-ai-uptake-in-the-workplace-20240205.html

McKinsey Global Institute. (2025). *The economic potential of generative AI*. Retrieved from https://www.mckinsey.com

SoftwareSeni. (2025). *What the research actually shows about AI coding assistant productivity*. Retrieved from https://www.softwareseni.com/what-the-research-actually-shows-about-ai-coding-assistant-productivity/

Tom's Hardware. (2025). *Workers save time using AI tools*. Retrieved from https://www.tomshardware.com/tech-industry/artificial-intelligence

Yaware. (2025). *The death of micromanagement*. Retrieved from https://yaware.com/blog/the-death-of-micromanagement-how-real-time-productivity-data-transforms-leadership/

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